Cost-Aware PAC Labeling from a Linear-Programming Perspective

STATS 357 Project

May 26, 2026

Abstract

We revisit cost-aware probably approximately correct (PAC) labeling through a linear-programming perspective. For a finite target pool, multiple cheap labeling sources, and access to an expert labeler, the oracle problem is to minimize labeling cost subject to an average-error constraint. The Lagrange dual of this LP gives a scalar shadow price for error, and the resulting itemwise minimizer is a fixed-price routing rule over cheap sources and expert deferral. Since the oracle losses are unavailable before labeling, the implementable method uses proxy losses only to generate candidate prices and policies. Finite-sample validity is then supplied by held-out calibration labels, which certify the actual loss of the selected policy without requiring the proxy to be correct. This separates optimization from inference: the LP and proxy model seek cost efficiency, while calibration provides the PAC certificate. We also describe an audited online extension in which randomized expert audits produce inverse-probability feedback for updating the dual price and controlling finite-horizon realized error.

1 A Linear-Programming View of Cost-Aware PAC Labeling

1.1 Setup and Notation

We preserve the online notation $t = 1, \dots, T$ so that the offline and online formulations remain compatible. There is a fixed target pool

$$\mathcal{T} = \{X_t\}_{t=1}^{T},$$

whose true labels $Y_1, \dots, Y_T$ are initially unknown. For each item t and each cheap source $j \in [k]$, we observe a predicted label

$$\widehat{Y}_t^{(j)}$$

and a known query cost $c_{t,j} \geq 0$. The expert label Y_t is treated as correct and costs $c_{t,E} > 0$. The loss is bounded:

$$0 \leq \ell\big(Y_t, \widehat{Y}_t^{(j)}\big) \leq 1, \qquad t \in [T],\ j \in [k]. \tag{1}$$

The expert action has zero labeling loss.

Define the action set

$$\mathcal{A} = \{E, 1, \dots, k\},$$

where E denotes deferral to the expert. For $a \in \mathcal{A}$, define the itemwise loss and cost

$$h_t(a) = \begin{cases} 0, & a = E, \\ \ell\big(Y_t, \widehat{Y}_t^{(a)}\big), & a \in [k], \end{cases} \qquad C_t(a) = \begin{cases} c_{t,E}, & a = E, \\ c_{t,a}, & a \in [k]. \end{cases} \tag{2}$$

A labeling rule chooses an action $a_t \in \mathcal{A}$ for each item. The resulting final label is

$$\widetilde{Y}_t = \begin{cases} Y_t, & a_t = E, \\ \widehat{Y}_t^{(a_t)}, & a_t \in [k]. \end{cases}$$

The finite-pool average labeling error and average cost are

$$L_T(a_{1:T}) = \frac{1}{T}\sum_{t=1}^{T} h_t(a_t), \qquad C_T(a_{1:T}) = \frac{1}{T}\sum_{t=1}^{T} C_t(a_t). \tag{3}$$

The offline target is

$$\min_{a_1,\dots,a_T\in\mathcal{A}} C_T(a_{1:T}) \qquad \text{subject to} \qquad L_T(a_{1:T}) \le \varepsilon. \tag{4}$$

1.2 The Oracle Cost-Aware Labeling LP

The deterministic problem in (4) is the cost-aware labeling problem we would like to solve. To expose the pricing structure, first consider its randomized LP relaxation. Let $x_{t,a}$ be the probability of choosing action a on item t. If the true losses $h_t(a)$ were known for all target items and all actions, the oracle policy would solve

$$\text{minimize}_x \quad \frac{1}{T}\sum_{t=1}^{T}\sum_{a\in\mathcal{A}} C_t(a)x_{t,a} \tag{5}$$

$$\text{subject to} \quad \frac{1}{T}\sum_{t=1}^{T}\sum_{a\in\mathcal{A}} h_t(a)x_{t,a} \le \varepsilon, \tag{6}$$

$$\sum_{a\in\mathcal{A}} x_{t,a} = 1, \qquad t = 1,\dots,T, \tag{7}$$

$$x_{t,a} \ge 0, \qquad t\in[T],\ a\in\mathcal{A}.$$

The integer formulation is recovered by requiring $x_{t,a} \in \{0,1\}$. The LP relaxation is useful because its dual reveals the one-dimensional price that will later index our candidate routing policies.

1.3 Lagrangian Dual and Shadow Price

Introduce one dual variable $\lambda \ge 0$ for the average-error constraint and one free dual variable $\mu_t \in \mathbb{R}$ for each simplex equality. The nonnegativity constraints $x_{t,a} \ge 0$ are kept as the domain of the inner minimization.

Primal constraint	Dual variable	Sign restriction
$\frac{1}{T}\sum_{t,a} h_t(a)x_{t,a} \le \varepsilon$	λ	$\lambda \ge 0$
$\sum_a x_{t,a} = 1$ for item t	μ_t	$\mu_t \in \mathbb{R}$
$x_{t,a} \ge 0$	domain constraint	–

Using the equality convention $\mu_t(1 - \sum_a x_{t,a})$, the Lagrangian is

$$\begin{aligned}\mathcal{L}(x;\lambda,\mu) &= \frac{1}{T}\sum_{t,a} C_t(a)x_{t,a} + \lambda\left(\frac{1}{T}\sum_{t,a} h_t(a)x_{t,a} - \varepsilon\right) + \sum_{t=1}^{T}\mu_t\left(1 - \sum_a x_{t,a}\right) \\ &= \sum_{t,a}\left[\frac{1}{T}\{C_t(a) + \lambda h_t(a)\} - \mu_t\right]x_{t,a} + \sum_{t=1}^{T}\mu_t - \lambda\varepsilon. \end{aligned} \tag{8}$$

For fixed (λ, μ), the inner problem is $\inf_{x \geq 0} \mathcal{L}(x; \lambda, \mu)$. Since the Lagrangian is linear in every $x_{t,a}$, this infimum is finite if and only if

$$\mu_t \leq \frac{1}{T}\{C_t(a) + \lambda h_t(a)\}, \qquad t \in [T],\ a \in \mathcal{A}. \tag{9}$$

Otherwise some variable with a negative coefficient can be sent to $+\infty$, driving the Lagrangian to $-\infty$. When (9) holds, the x terms vanish at the inner minimum and the dual function is

$$g(\lambda, \mu) = \sum_{t=1}^{T} \mu_t - \lambda\varepsilon.$$

Thus the explicit dual LP is

$$\begin{aligned} \text{maximize}_{\lambda,\mu} \quad & \sum_{t=1}^{T} \mu_t - \lambda\varepsilon && (10)\\ \text{subject to} \quad & \mu_t \leq \frac{1}{T}\{C_t(a) + \lambda h_t(a)\}, \qquad t \in [T],\ a \in \mathcal{A}, \\ & \lambda \geq 0, \qquad \mu_t \in \mathbb{R},\ t \in [T]. \end{aligned}$$

The variables μ_t can be eliminated analytically. For any fixed λ, the objective is increasing in each μ_t, so the optimal value is attained at the largest feasible μ_t:

$$\mu_t^\star(\lambda) = \frac{1}{T} \min_{a \in \mathcal{A}} \{C_t(a) + \lambda h_t(a)\}.$$

Substitution gives the reduced one-dimensional dual

$$\max_{\lambda \geq 0} D(\lambda) := -\lambda\varepsilon + \frac{1}{T} \sum_{t=1}^{T} \min_{a \in \mathcal{A}} \{C_t(a) + \lambda h_t(a)\}. \tag{11}$$

The function $D(\lambda)$ is concave because it is a sum of pointwise minima of affine functions of λ, minus a linear term.

This is the source of the fixed-price rule: once a price λ is fixed, the dual minimization separates across items, and item t chooses an action in

$$a_t(\lambda) \in \arg\min_{a \in \mathcal{A}} \{C_t(a) + \lambda h_t(a)\}. \tag{12}$$

The price λ is the shadow price of the single average-error constraint: small λ makes the rule cost-seeking, while large λ makes it error-averse.

Proposition 1 (Dual problem and fixed-price routing)**.** *The Lagrange dual of* (5)–(7) *is the explicit dual* (10)*, equivalently the reduced dual* (11)*. Strong duality holds. Moreover, for any dual optimal price $\lambda^\star$, any primal optimizer can be chosen to put positive mass only on actions attaining the minimum in* (12)*. If the deterministic fixed-price policy $a_t(\lambda^\star)$ is feasible and satisfies complementary slackness,*

$$\lambda^\star\{L_T(a_{1:T}(\lambda^\star)) - \varepsilon\} = 0,$$

then it is an optimal deterministic solution.

Lemma 1 (At most one fractional item)**.** *There exists an extreme-point solution of the oracle LP in which all but at most one item t are assigned deterministically. Consequently, the gap between the randomized LP and a deterministic policy obtained by rounding the one fractional item is at most $1/T$ in average loss and at most $\max_{t,a} C_t(a)/T$ in average cost.*

Everything in this section is an oracle calculation: the losses $h_t(a)$ are treated as known so that the cost-aware LP and its dual can be written exactly. In the deployable problem, the cheap-source losses are precisely the unknown quantities. The next section replaces h_t by an estimated proxy $\widehat{h}_t$ and then uses calibration, rather than proxy correctness, to certify the realized error.

2 Offline PAC Labeling via Calibration

2.1 Offline Proxy Prices

The true cheap-source losses $h_t(j)$ are not known before expert labels are collected. The offline development stage therefore learns or estimates a proxy

$$\widehat{h}_{t,j} = \widehat{h}_j(X_t, \widehat{Y}_t^{(j)}, \text{metadata}) \in [0,1] \tag{13}$$

for the loss of cheap source j on item t. The proxy may be obtained from a labeled training split by regression, isotonic calibration, multicalibration, or any other supervised procedure. The certification theorem below does not require $\widehat{h}_{t,j}$ to be conservative or well calibrated. Better proxies reduce cost; held-out calibration supplies validity.

Set $\widehat{h}_t(E) = 0$ and $\widehat{h}_t(j) = \widehat{h}_{t,j}$ for $j \in [k]$. The implementable rule is the plug-in version of (12):

$$a_t^\lambda \in \arg\min_{a \in \mathcal{A}} \{C_t(a) + \lambda \widehat{h}_t(a)\}.$$

Equivalently, first choose the cheapest proxy-adjusted cheap source,

$$j_t^\lambda \in \arg\min_{j \in [k]} \{c_{t,j} + \lambda \widehat{h}_{t,j}\}, \tag{14}$$

and then defer to the expert when the expert is no more expensive than the selected cheap source after pricing the predicted loss:

$$D_t^\lambda = \mathbf{1}\{c_{t,E} \le c_{t,j_t^\lambda} + \lambda \widehat{h}_{t,j_t^\lambda}\}. \tag{15}$$

Here $D_t^\lambda = 1$ means deferral. The action is

$$a_t^\lambda = \begin{cases} E, & D_t^\lambda = 1, \\ j_t^\lambda, & D_t^\lambda = 0. \end{cases} \tag{16}$$

Remark 1 (Relation to the single-model PAC threshold)**.** *Once a candidate price $\lambda \ge 0$ is fixed, the rule above reduces to the usual PAC-labeling threshold rule in the single-source special case. Indeed, with one cheap source, zero cheap-source cost, constant expert cost c_E, and a scalar proxy $\widehat{h}_t$ for cheap-source loss, the AI label is used exactly when*

$$\lambda \widehat{h}_t < c_E, \qquad \textit{equivalently} \qquad \widehat{h}_t < c_E/\lambda$$

for $\lambda > 0$. Thus the usual uncertainty threshold is the cost-aware threshold c_E/λ. In the proposed algorithm, λ indexes a candidate policy and is selected from calibration data in Section 2.2; it is not assumed known in advance. With multiple cheap sources, this scalar threshold is replaced by the itemwise minimum of cheap-source costs plus priced predicted losses.

The proxy LP viewpoint is useful but not sufficient for validity. For example, replacing h_t by $\widehat{h}_t$ in (6) would produce a proxy-constrained LP, and the rule above is exactly its Lagrangian minimizer. However, the proxy constraint is not an error certificate unless the proxy is known to dominate the true loss. The role of the PAC calibration step is to certify the actual loss of the candidate prices generated by this plug-in dual rule.

The counterfactual AI-decision loss of a fixed proxy-price policy is

$$H_t(\lambda) = (1 - D_t^\lambda)\ell\big(Y_t, \widehat{Y}_t^{(j_t^\lambda)}\big), \tag{17}$$

and its deployment cost is

$$C_t(\lambda) = D_t^\lambda c_{t,E} + (1 - D_t^\lambda)c_{t,j_t^\lambda}. \tag{18}$$

The finite-pool averages are

$$L_T(\lambda) = \frac{1}{T}\sum_{t=1}^{T} H_t(\lambda), \qquad C_T(\lambda) = \frac{1}{T}\sum_{t=1}^{T} C_t(\lambda). \tag{19}$$

2.2 Offline Calibration and Price Selection

Let Λ be a finite candidate price set. It may be a user-specified grid, or the set of policies induced by all breakpoints of the proxy-adjusted costs; Appendix D records this finite reduction. The candidate set may depend on the target covariates, cheap-source predictions, costs, and separate training data used to learn $\widehat{h}$, but it must not depend on the calibration labels used below.

For a labeled calibration sample $I_1, \dots, I_m$, define

$$\widehat{L}_m(\lambda) = \frac{1}{m}\sum_{s=1}^{m} H_{I_s}(\lambda). \tag{20}$$

The uniform Hoeffding upper confidence bound is

$$U_m(\lambda;\alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{\log(|\Lambda|/\alpha)}{2m}}. \tag{21}$$

The certified price is selected by

$$\hat{\lambda} \in \operatorname*{arg\,min}_{\lambda\in\Lambda} \left\{C_T(\lambda) : U_m(\lambda;\alpha) \le \varepsilon\right\}. \tag{22}$$

If the feasible set is empty, include the expert-only policy as a candidate, denoted $\lambda = \infty$, for which $D_t^\infty = 1$ and $L_T(\infty) = 0$.

2.3 Offline PAC Guarantees

2.3.1 Transductive Finite-Pool Guarantee

The transductive setting is closest to fixed-dataset PAC labeling. The target pool $\mathcal{T}$ is fixed, and calibration indices $I_1, \dots, I_m$ are drawn independently from $\mathrm{Unif}([T])$. The expert labels Y_{I_s} are then revealed. For each fixed λ,

$$\mathbb{E}\big[H_{I_s}(\lambda) \mid \mathcal{T}, Y_{1:T}\big] = L_T(\lambda).$$

Algorithm 1 Offline fixed-price cost-aware PAC labeling

Require: Target pool $\{X_t\}_{t=1}^T$; cheap predictions $\{\widehat{Y}_t^{(j)}\}_{t,j}$; costs $\{c_{t,j}\}_{t,j}$ and $\{c_{t,E}\}_t$; target error ε; failure level α; candidate prices Λ.

Ensure: A certified price $\hat{\lambda}$ and final labels $\widetilde{Y}_1, \dots, \widetilde{Y}_T$.

1: Use a labeled training split, if available, to fit proxy losses $\widehat{h}_{t,j} \in [0, 1]$.
2: For every $\lambda \in \Lambda$ and target item t, compute j_t^λ, D_t^λ, and $C_t(\lambda)$ using (14)–(18).
3: On an independent labeled calibration split, evaluate $H_{I_s}(\lambda)$ and $\widehat{L}_m(\lambda)$ for every $\lambda \in \Lambda$.
4: Compute $U_m(\lambda; \alpha)$ from (21).
5: Select $\hat{\lambda}$ by (22); fall back to the expert-only policy if no nontrivial price is certified.
6: For each target item, output the expert label if $D_t^{\hat{\lambda}} = 1$; otherwise output $\widehat{Y}_t^{(j_t^{\hat{\lambda}})}$.
7: If calibration items belong to the target pool, use their already observed expert labels in the released dataset.

Theorem 1 (Offline fixed-price PAC guarantee)**.** *Assume* (1)*. Let Λ be finite and independent of the calibration labels conditional on the target pool, cheap-source predictions, costs, and any separate training data used to construct $\widehat{h}$. Let $\hat{\lambda}$ be selected by* (22) *using* (21)*. Then, with probability at least $1 - \alpha$ over the random draw of the calibration indices,*

$$L_T(\hat{\lambda}) \le \varepsilon. \tag{23}$$

Consequently, the labels produced by the fixed-price policy $a_t^{\hat{\lambda}}$ satisfy

$$\frac{1}{T} \sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \le \varepsilon \tag{24}$$

with probability at least $1 - \alpha$. If calibration items are corrected to expert labels in the final dataset, the actual final loss is no larger than $L_T(\hat{\lambda})$.

The proof is in Appendix A. The key point is uniformity. If a price were fixed before observing calibration labels, a one-price confidence bound would suffice. Because the algorithm selects the cheapest empirically certified price, the confidence event must hold for every candidate policy in Λ.

2.3.2 Inductive Deployment Guarantee

Corollary 1 (Inductive deployment guarantee)**.** *Suppose the calibration set is an independent labeled sample from a distribution P, and the deployment items are drawn independently from the same distribution. For a price λ, define the population risk*

$$L(\lambda) = \mathbb{E}_{(X,Y)\sim P} \left[(1 - D^\lambda(X)) \ell\big(Y, \widehat{Y}^{(j^\lambda(X))}(X)\big) \right].$$

Assume Λ is finite and independent of the calibration labels conditional on covariates, cheap-source predictions, costs, and any separate training data used to construct $\widehat{h}$. Let $\hat{\lambda}$ be selected by (22) *using the calibration sample. Then $L(\hat{\lambda}) \le \varepsilon$ with probability at least $1 - \alpha$. For a finite deployment pool of size T, with probability at least $1 - \alpha - \delta$,*

$$\frac{1}{T} \sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \le \varepsilon + \sqrt{\frac{\log(1/\delta)}{2T}}. \tag{25}$$

If the desired realized deployment-pool average is at most ε, the internal certification target should be reduced by the final square-root term.

3 Online PAC Labeling via Audits

The offline guarantee certifies a fixed policy after a labeled calibration stage. In an online data-curation problem, inputs arrive sequentially, decisions are made before future data are seen, and the loss of an AI-labeled item is unobserved unless an expert label is revealed. Therefore the offline calibration argument cannot be reused directly. The natural online analogue is to keep the dual-price decision rule but add randomized audits of AI-labeled points.

3.1 Online Streaming Setup and Audited Feedback

At time t, the learner observes X_t, cheap labels $\widehat{Y}_t^{(j)}$, costs $c_{t,j}$, expert cost $c_{t,E}$, and predictable proxy losses $\widehat{h}_{t,j}$. The price λ_t is chosen from past information. The online fixed-price decision is

$$j_t^\star \in \arg\min_{j\in[k]}\{c_{t,j} + \lambda_t \widehat{h}_{t,j}\}, \tag{26}$$

followed by expert deferral if

$$c_{t,E} \le c_{t,j_t^\star} + \lambda_t \widehat{h}_{t,j_t^\star}. \tag{27}$$

Let $D_t = 1$ denote this final-label deferral decision. If $D_t = 1$, the output label is Y_t and the final loss is zero. If $D_t = 0$, the learner initially outputs $\widehat{Y}_t^{(j_t^\star)}$. It then draws an independent audit variable

$$A_t \sim \mathrm{Bernoulli}(p_t), \qquad p_t \in [p_{\min}, 1].$$

If $A_t = 1$, the expert label is revealed and the final released label may be corrected to Y_t. The audit still records the counterfactual AI error for updating the price.

Define the counterfactual AI-decision loss

$$H_t = (1 - D_t)\ell\big(Y_t, \widehat{Y}_t^{(j_t^\star)}\big),$$

where deferral has zero loss because the expert label is used. Deferral is an observed zero-loss event, while an unaudited AI decision has unobserved loss. Define the augmented observation mask and its conditional observation probability by

$$O_t = D_t + (1 - D_t)A_t, \qquad q_t = D_t + (1 - D_t)p_t. \tag{28}$$

Thus $q_t = 1$ when $D_t = 1$ and $q_t = p_t$ when $D_t = 0$. The centered inverse-observation update signal is

$$\widehat{Z}_t = \frac{O_t}{q_t}\,(H_t - \varepsilon_0). \tag{29}$$

The realized final loss after audit-and-correct is

$$G_t = \ell(Y_t, \widetilde{Y}_t), \qquad G_t \le H_t.$$

The online price update is

$$\lambda_{t+1} = [\lambda_t + \eta \widehat{Z}_t]_+, \tag{30}$$

where $\eta > 0$ is a step size and $\varepsilon_0 \le \varepsilon$ is an internal error target.

Algorithm 2 Audited online dual-price PAC labeling

Require: Stream horizon T; target error ε; audit floor $p_{\min}$; step size η; internal target ε_0.

```
Initialize λ_1 ≥ 0.
for t = 1, ..., T do
    Observe X_t, cheap labels {Ŷ_t^(j)}_{j=1}^k, costs, and predictable proxies {ĥ_{t,j}}_{j=1}^k.
    Choose j_t^⋆ by (26).
    if (27) holds then
        Query the expert, output Ỹ_t = Y_t, set D_t = 1 and Ẑ_t = −ε_0.
    else
        Output the AI label Ỹ_t = Ŷ_t^(j_t^⋆) and set D_t = 0.
        Draw A_t ∼ Bernoulli(p_t) with p_t ∈ [p_min, 1].
        if A_t = 1 then
            Query the expert, optionally correct Ỹ_t = Y_t, and set Ẑ_t = {ℓ(Y_t, Ŷ_t^(j_t^⋆)) − ε_0}/p_t.
        else
            Set Ẑ_t = 0.
        end if
    end if
    Update λ_{t+1} by (30).
end for
```

3.2 Online Finite-Horizon Guarantee

The online result is distribution-free with respect to the data sequence; the probability is over the algorithm's audit randomness. It controls realized error, not cost regret against the offline LP optimum.

Assumption 1 (Predictable routing and random audits)**.** *The routing decision, audit probability, and selected source are chosen using only past information and the current non-expert information* $(X_t, \{\widehat{Y}_t^{(j)}, \widehat{h}_{t,j}, c_{t,j}\}_{j=1}^k, c_{t,E})$*. After this decision, the audit draw is independent Bernoulli with known probability* $p_t \in [p_{\min}, 1]$*.*

Assumption 2 (Proxy floor and bounded expert cost)**.** *For some* $\rho > 0$*,* $\widehat{h}_{t,j} \in [\rho, 1]$ *for all* t, j*. For the fixed horizon, let*

$$c_{\min} = \min_{t \le T,\, j \in [k]} c_{t,j}, \qquad c_E^{\max} = \max_{t \le T} c_{t,E}.$$

If future costs are not known in advance, these extrema may be replaced by any deterministic lower bound on cheap-source costs and upper bound on expert costs that are known before deployment.

Proposition 2 (Unbiased intermittent update signal)**.** *Under bounded loss and online predictability,*

$$\mathbb{E}\left[\frac{O_t}{q_t} \,\middle|\, \mathcal{G}_t\right] = 1, \qquad \mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = H_t - \varepsilon_0.$$

Theorem 2 (Audited online finite-horizon control)**.** *Assume bounded loss, online predictability and audits, and the proxy floor. Let*

$$\lambda_{\mathrm{def}} = \frac{(c_E^{\max} - c_{\min})_+}{\rho}, \qquad \Lambda_{\mathrm{on}} = \max\left\{\lambda_1, \lambda_{\mathrm{def}} + \frac{\eta(1 - \varepsilon_0)}{p_{\min}}\right\}.$$

Run Algorithm 2 with constant step size $\eta > 0$ and internal target $\varepsilon_0 \in [0, \varepsilon]$. Then, for any fixed horizon T and any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the audit randomness,

$$\frac{1}{T}\sum_{t=1}^{T} H_t \leq \varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}}. \tag{31}$$

Consequently, if audited AI labels are corrected to expert labels in the final dataset, then

$$\frac{1}{T}\sum_{t=1}^{T} G_t \leq \varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}}. \tag{32}$$

Corollary 2 (PAC-style online tuning)**.** *Under the conditions of Theorem 2, suppose*

$$\varepsilon_0 + \frac{\Lambda_{\mathrm{on}}}{\eta T} + \frac{1}{p_{\min}}\sqrt{\frac{2\log(1/\delta)}{T}} \leq \varepsilon,$$

with $\varepsilon_0 \geq 0$. Then the audit-and-correct final labels satisfy

$$\frac{1}{T}\sum_{t=1}^{T} G_t \leq \varepsilon$$

with probability at least $1 - \delta$. With $\eta \asymp T^{-1/2}$ and constant $p_{\min}$, the slack is $O(T^{-1/2})$.

3.3 Offline versus Online Certification

The online theorem is deliberately different from the offline one. Offline calibration certifies a fixed candidate policy by evaluating it on a held-out labeled sample. Online auditing controls the long-run error of a moving policy by converting unobserved AI losses into an unbiased intermittent-feedback signal. The proxy losses again affect efficiency rather than validity, but in the online setting some audit or reveal mechanism is essential: without it, two streams can induce identical observations while having opposite AI-label correctness on every unaudited item.

4 Experiments

We run a small proof-of-concept study on ImageNet and ImageNetV2 using the confidence score already included in the data. There is one cheap source, namely the model prediction, and one expert source, namely the ground-truth label. The cheap-source proxy loss is $\widehat{h} = 1 - \text{confidence}$, the cheap source has zero marginal cost, and the expert has unit cost. All reported numbers average over 500 random trials. Appendix B gives the full sweep and implementation details.

Offline calibration. The offline experiment compares the fixed-price rule from Section 2 with the original scalar PAC threshold baseline. Because this experiment has only one cheap source, the fixed-price policies are nested uncertainty-threshold rules. We therefore certify the fixed-price rule with the sharper monotone confidence bound

$$\hat{L}_m(\lambda) + \sqrt{\frac{\log(1/\alpha)}{2m}},$$

rather than the finite-class union bound used for general multi-source policy classes. At $\varepsilon = 0.10$, both methods have zero violations in 500 trials and nearly identical expert-budget savings.

Dataset	Method	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.0873	0.0000	0.7848
ImageNet	original PAC	0.0879	0.0000	0.7864
ImageNetV2	fixed-price monotone UCB	0.0718	0.0000	0.5457
ImageNetV2	original PAC	0.0726	0.0000	0.5479

Table 1: Offline results at $\varepsilon = 0.10$. "Budget saved" is one minus the fraction of items sent to the expert.

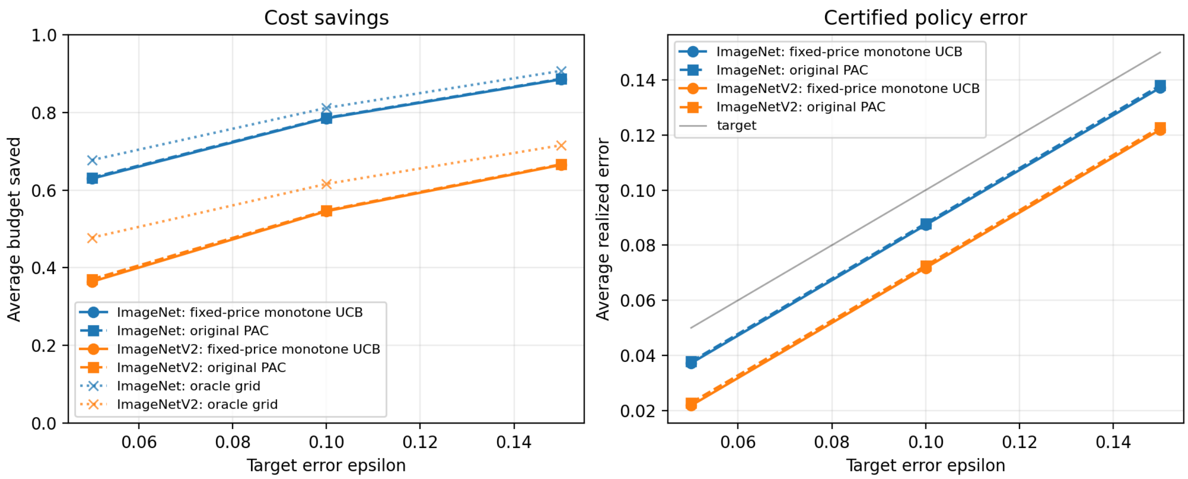

Figure 1: Offline sweep over target error levels. After exploiting the single-source monotonicity, the fixed-price rule is nearly indistinguishable from the original scalar PAC baseline; the remaining gap comes from using a fixed threshold grid rather than the exact calibration-order threshold.

Online audited routing. The online experiment uses a random stream order and the dual-price update in Section 3 with audit probabilities $p \in \{0.1, 0.2, 0.5\}$. The update follows the intermittent-feedback form: an unaudited AI decision does not move the price, while an expert deferral is treated as an observed zero-loss event. These runs set the internal online target to $\varepsilon_0 = \varepsilon$, so they are a behavioral check of the audited update rather than a theorem-tight slack-adjusted implementation of Corollary 2. Lower audit rates save more expert budget but leave less correction, so final error is closer to the target. At $\varepsilon = 0.10$, ImageNet has no final-error violations for all three audit rates; ImageNetV2 has a small violation rate at $p = 0.1$ and none at $p \geq 0.2$.

Dataset	p	Final error	AI error	Violation	Budget saved
ImageNet	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.5	0.0490	0.0981	0.0000	0.3952
ImageNetV2	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.5	0.0499	0.0996	0.0000	0.3010

Table 2: Online audited results at $\varepsilon = 0.10$. "AI error" is the counterfactual average error of AI-labeled items before audit correction; "Final error" is after audited items are corrected by the expert.

5 Related Work

PAC labels and conformal calibration. The closest starting point is PAC labeling [1], which gives finite-sample guarantees for cost-effective dataset labeling using AI predictions and expert audits. That framework already includes both scalar-threshold PAC labeling and multi-model PAC routing variants. Our formulation is best viewed as a complementary optimization-first presentation: we start from a finite-pool cost-minimization LP, derive the fixed-price routing rule from its dual, allow item- and source-specific costs, and then use calibration only to certify the actual loss of the selected proxy-generated policy. The calibration logic is closely related to conformal prediction and distribution-free uncertainty quantification [2, 3], but the object being controlled here is the average error of a released labeled dataset rather than coverage of prediction sets.

Online conformal prediction with partial feedback. Adaptive and online conformal methods update calibration parameters over time to handle changing environments [4]. Recent work has studied intermittent or semi-bandit feedback, where labels are not always observed [5, 6, 7]. Those papers motivate the audited online extension in Section 3: when labels are not always revealed, the feedback used to update a calibration or price parameter must be corrected for the observation process.

Learning to defer and cost-sensitive routing. Learning-to-defer methods train systems that either predict or pass an item to an expert [8, 9]. Multiple-expert and cost-sensitive variants study richer deferral and routing settings [10, 11]. The distinction here is the finite-sample PAC certificate for the final released labels. The proxy router may be learned by standard supervised methods, but the guarantee comes from held-out calibration of the actual selected policy.

6 Discussion

The report separates two regimes. The offline method has a clean division of labor. The proxy model and the LP structure make the policy cost efficient; the calibration set makes it valid. When instantiating the method, the main modeling choice is whether the target is transductive or inductive. If the application is dataset curation for a fixed unlabeled pool, the transductive theorem should be the main result. If the application is repeated deployment on future items, the inductive statement should be foregrounded. The online method addresses a different obstacle: the selected policy changes over time and AI losses are not observed without expert reveal. Random audits supply the missing feedback through an inverse-probability loss estimate. This gives finite-horizon error control, but not a cost-regret comparison to the offline oracle LP. The experiments in Section 4 illustrate this distinction: offline calibration gives a clean finite-policy certificate, while the online mechanism exposes an explicit audit-cost versus correction tradeoff. The main empirical limitation is that the current data contain only one cheap source. A stronger test of the LP perspective would use several cheap sources with heterogeneous costs, where the original scalar threshold structure no longer captures the routing problem.

A Proofs

Proof of Proposition 1. The derivation in Section 1 gives the Lagrangian (8). Taking the infimum over $x_{t,a} \geq 0$ yields the dual feasibility constraints (9) and hence the explicit dual (10). Eliminating μ_t gives (11). The primal is feasible because choosing the expert for all items gives zero loss, and it is bounded because costs are nonnegative and finite. Therefore strong duality holds by finite-dimensional linear programming duality. For the support statement, complementary slackness for the primal variable $x_{t,a}$ implies that $x_{t,a} > 0$ only when

$$\mu_t = \frac{1}{T}\{C_t(a) + \lambda h_t(a)\}.$$

At a dual optimum, μ_t is the minimum feasible upper bound over actions, so this equality can hold only for actions minimizing $C_t(a) + \lambda h_t(a)$. The deterministic optimality claim is then the standard complementary-slackness condition for the global risk constraint. □

Proof of Lemma 1. At an extreme point, the number of positive variables is at most the number of linearly independent active equality constraints after converting the active global risk constraint, if any, to an equality. There are T row-sum constraints and at most one active global risk constraint, so an extreme point has at most $T+1$ positive variables. Every row t must have at least one positive variable because $\sum_a x_{t,a} = 1$. Thus at most one row can contain more than one positive variable. Rounding only that row changes one term in the loss average and one term in the cost average, giving the stated bounds. □

Proof of Theorem 1. For each fixed $\lambda \in \Lambda$, the variables $H_{I_s}(\lambda)$ are i.i.d. in $[0,1]$ with mean $L_T(\lambda)$ conditional on the target pool. Hoeffding's inequality gives

$$\mathbb{P}\left\{L_T(\lambda) > \widehat{L}_m(\lambda) + \sqrt{\frac{\log(|\Lambda|/\alpha)}{2m}}\right\} \leq \frac{\alpha}{|\Lambda|}.$$

A union bound over Λ implies that, with probability at least $1-\alpha$,

$$L_T(\lambda) \leq U_m(\lambda;\alpha) \qquad \text{for every } \lambda \in \Lambda.$$

On this event, the selected price satisfies

$$L_T(\hat{\lambda}) \leq U_m(\hat{\lambda};\alpha) \leq \varepsilon$$

by construction. The final label loss equals $H_t(\hat{\lambda})$ on items not corrected by the expert and is at most $H_t(\hat{\lambda})$ on calibration items if they are released with their expert labels. This proves the claim. □

Proof of Corollary 1. For each fixed $\lambda \in \Lambda$, the calibration losses are i.i.d. in $[0,1]$ with mean $L(\lambda)$. Hoeffding's inequality and a union bound over Λ imply that, with probability at least $1-\alpha$ over the calibration sample,

$$L(\lambda) \leq U_m(\lambda;\alpha) \qquad \text{for every } \lambda \in \Lambda.$$

On this event, the selected price satisfies $L(\hat{\lambda}) \leq U_m(\hat{\lambda};\alpha) \leq \varepsilon$. Conditional on the calibration sample and on this event, the deployment losses $\ell(Y_t, \widetilde{Y}_t)$ are independent, bounded in $[0,1]$, and have conditional mean at most $L(\hat{\lambda}) \leq \varepsilon$. Hoeffding's inequality therefore gives

$$\frac{1}{T}\sum_{t=1}^{T} \ell(Y_t, \widetilde{Y}_t) \leq \varepsilon + \sqrt{\frac{\log(1/\delta)}{2T}}$$

with conditional probability at least $1-\delta$. Combining the calibration and deployment events gives probability at least $1-\alpha-\delta$. □

Proof of Proposition 2. For the analysis, let $\mathcal{G}_t$ denote the post-decision sigma-field that fixes the selected source, audit probability, deferral decision, and the latent selected loss. The learner need not observe the latent loss; it is included only so the conditional expectation is well defined. If $D_t = 1$, then $O_t = q_t = 1$ and $H_t = 0$, so

$$\mathbb{E}\left[\frac{O_t}{q_t}\,\middle|\,\mathcal{G}_t\right] = 1, \qquad \mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = -\varepsilon_0 = H_t - \varepsilon_0.$$

If $D_t = 0$, then $O_t = A_t$, $q_t = p_t$, and conditional on $\mathcal{G}_t$ the audit draw is Bernoulli(p_t) while the loss $\ell(Y_t, \widehat{Y}_t^{(j_t^\star)})$ is fixed. Hence

$$\mathbb{E}\left[\frac{O_t}{q_t}\,\middle|\,\mathcal{G}_t\right] = \frac{p_t}{p_t} = 1.$$

Since H_t and ε_0 are fixed conditional on $\mathcal{G}_t$, this also gives

$$\mathbb{E}[\widehat{Z}_t \mid \mathcal{G}_t] = \mathbb{E}\left[\frac{O_t}{q_t}(H_t - \varepsilon_0)\,\middle|\,\mathcal{G}_t\right] = H_t - \varepsilon_0.$$

□

Proof of Theorem 2. First bound the price process. If $\lambda_t \geq \lambda_{\mathrm{def}}$, then for every cheap source j,

$$c_{t,j} + \lambda_t \widehat{h}_{t,j} \geq c_{\min} + \lambda_{\mathrm{def}}\rho \geq c_E^{\max} \geq c_{t,E}.$$

Thus the algorithm defers to the expert, so $D_t = 1$ and $\widehat{Z}_t = -\varepsilon_0$. Consequently $\lambda_{t+1} = [\lambda_t - \eta\varepsilon_0]_+ \leq \lambda_t$. If instead $\lambda_t < \lambda_{\mathrm{def}}$, then $\widehat{Z}_t \leq (1-\varepsilon_0)/p_{\min}$ and hence

$$\lambda_{t+1} \leq \lambda_t + \eta(1-\varepsilon_0)/p_{\min} < \lambda_{\mathrm{def}} + \eta(1-\varepsilon_0)/p_{\min}.$$

By induction, $0 \leq \lambda_t \leq \Lambda_{\mathrm{on}}$ for all t.

Since $[x]_+ \geq x$, the update gives

$$\lambda_{t+1} \geq \lambda_t + \eta\widehat{Z}_t.$$

Summing over $t = 1, \ldots, T$ yields

$$\sum_{t=1}^{T} \widehat{Z}_t \leq \frac{\lambda_{T+1} - \lambda_1}{\eta} \leq \frac{\Lambda_{\mathrm{on}}}{\eta}.$$

Let $M_t = (H_t - \varepsilon_0) - \widehat{Z}_t$. By Proposition 2, M_t is a martingale difference with respect to the audit filtration. Moreover, $|M_t| \leq 1/p_{\min}$. Azuma-Hoeffding gives, with probability at least $1-\delta$,

$$\sum_{t=1}^{T} M_t \leq \frac{1}{p_{\min}}\sqrt{2T\log(1/\delta)}.$$

Because

$$\sum_{t=1}^{T} H_t = T\varepsilon_0 + \sum_{t=1}^{T} \widehat{Z}_t + \sum_{t=1}^{T} M_t,$$

combining the last displays gives (31). Finally, audit-and-correct gives $G_t \leq H_t$ pointwise, which proves (32). □

Dataset	Method	ε	Error	Violation	Budget saved
ImageNet	fixed-price monotone UCB	0.05	0.0372	0.0000	0.6296
ImageNet	fixed-price monotone UCB	0.10	0.0873	0.0000	0.7848
ImageNet	fixed-price monotone UCB	0.15	0.1372	0.0000	0.8851
ImageNet	original PAC	0.05	0.0377	0.0000	0.6323
ImageNet	original PAC	0.10	0.0879	0.0000	0.7864
ImageNet	original PAC	0.15	0.1380	0.0000	0.8866
ImageNetV2	fixed-price monotone UCB	0.05	0.0218	0.0000	0.3642
ImageNetV2	fixed-price monotone UCB	0.10	0.0718	0.0000	0.5457
ImageNetV2	fixed-price monotone UCB	0.15	0.1219	0.0000	0.6648
ImageNetV2	original PAC	0.05	0.0228	0.0000	0.3706
ImageNetV2	original PAC	0.10	0.0726	0.0000	0.5479
ImageNetV2	original PAC	0.15	0.1228	0.0000	0.6665

Table 3: Full offline sweep. Error, violation, and budget saved are averages over 500 trials.

Proof of Corollary 2. The stated condition makes the right-hand side of (32) at most ε. If $\eta \asymp T^{-1/2}$, $p_{\min}$ is constant, and the costs and proxy floor are fixed so that $\Lambda_{\text{on}} = O(1)$, then both excess terms in (32) are $O(T^{-1/2})$. $\square$

B Experimental Details

The experiments use the two ImageNet CSV files from the HW4 copy directory: ImageNet has $T = 50000$ items and ImageNetV2 has $T = 10000$ items. Each file contains a ground-truth label, a model prediction, and a confidence score. We treat the model prediction as the only cheap source and the ground-truth label as the expert. The proxy loss is $\widehat{h} = 1 - \text{confidence}$, the cheap-source deployment cost is zero, and the expert cost is one. All experiments use 500 random trials. The scripts are `run_simple_offline_pac_experiment.py` and `run_simple_online_pac_experiment.py`.

Offline implementation. For the fixed-price method, the candidate threshold grid contains the expert-only rule plus 201 evenly spaced confidence-loss thresholds between 0 and 1. Although Section 2.2 states the general finite-policy bound, this single-source experiment uses the sharper monotone threshold radius

$$\sqrt{\frac{\log(1/\alpha)}{2m}},$$

which is valid uniformly over all scalar thresholds by the same nested-class argument used in the original PAC baseline. For each trial, the calibration indices are sampled independently and uniformly from the target pool with replacement, with calibration size $m = 0.2T$. The confidence level is $\alpha = 0.05$ and $\varepsilon \in \{0.05, 0.10, 0.15\}$. The original PAC baseline uses the same calibration sample and the same monotone radius, but chooses its threshold directly from the calibration mistake order instead of from the fixed grid.

Online implementation. For each trial, the target pool is randomly permuted into a stream. The update uses $\eta = 1$, proxy floor $\rho = 0.01$, and $\lambda_1 = 1/\varepsilon$. The online routing rule uses the AI prediction when $\lambda_t \max(1 - \text{confidence}_t, \rho) < 1$ and otherwise defers to the expert. If the AI

Dataset	ε	p	Final error	AI error	Violation	Budget saved
ImageNet	0.05	0.1	0.0436	0.0485	0.0100	0.5738
ImageNet	0.05	0.2	0.0394	0.0493	0.0000	0.5244
ImageNet	0.05	0.5	0.0249	0.0497	0.0000	0.3344
ImageNet	0.10	0.1	0.0780	0.0866	0.0000	0.6629
ImageNet	0.10	0.2	0.0737	0.0921	0.0000	0.6115
ImageNet	0.10	0.5	0.0490	0.0981	0.0000	0.3952
ImageNet	0.15	0.1	0.1017	0.1130	0.0000	0.7159
ImageNet	0.15	0.2	0.0979	0.1223	0.0000	0.6626
ImageNet	0.15	0.5	0.0683	0.1367	0.0000	0.4330
ImageNetV2	0.05	0.1	0.0445	0.0494	0.1840	0.4140
ImageNetV2	0.05	0.2	0.0393	0.0491	0.0100	0.3750
ImageNetV2	0.05	0.5	0.0244	0.0489	0.0000	0.2372
ImageNetV2	0.10	0.1	0.0863	0.0958	0.0400	0.5091
ImageNetV2	0.10	0.2	0.0783	0.0979	0.0000	0.4681
ImageNetV2	0.10	0.5	0.0499	0.0996	0.0000	0.3010
ImageNetV2	0.15	0.1	0.1197	0.1331	0.0000	0.5731
ImageNetV2	0.15	0.2	0.1126	0.1407	0.0000	0.5330
ImageNetV2	0.15	0.5	0.0739	0.1479	0.0000	0.3465

Table 4: Full online sweep. Final error, AI error, and budget saved are averages over 500 trials.

prediction is used, it is audited with probability $p \in \{0.1, 0.2, 0.5\}$. The hybrid intermittent update is $-\eta\varepsilon_0$ on expert deferrals, $\eta(\ell_t - \varepsilon_0)/p$ on audited AI decisions, and zero on unaudited AI decisions. Audited AI labels are corrected by the expert before release; unaudited AI labels are released as-is. The reported "AI error" is the counterfactual average loss before audit correction, and the reported final error is the realized loss after correction. As noted in Section 4, these runs set $\varepsilon_0 = \varepsilon$ rather than subtracting the finite-horizon slack from Corollary 2.

C Alternative Confidence Bounds

The Hoeffding bound (21) is simple but may be conservative. Any valid one-sided mean upper confidence bound can be substituted. For example, for a fixed λ, an empirical Bernstein bound gives

$$U_m^{\mathrm{EB}}(\lambda;\alpha) = \widehat{L}_m(\lambda) + \sqrt{\frac{2\hat{V}_m(\lambda)\log(2/\alpha)}{m}} + \frac{7\log(2/\alpha)}{3(m-1)}, \tag{33}$$

where $\hat{V}_m(\lambda)$ is the sample variance of $H_{I_s}(\lambda)$. For a finite grid, replace α by $\alpha/|\Lambda|$. Betting-based confidence sequences can also be used if the calibration stream is reused while expanding the candidate price set.

D Not Very Meaningful Results

The following observation is included only to show that a continuum of prices can be reduced to a finite policy class on a fixed finite dataset. It is not a main contribution: the resulting confidence penalty may be larger than that of a deliberately chosen price grid, and it does not remove the uniform-certification cost.

D.1 Continuous Prices and Breakpoint Enumeration

For each item t, set $\widehat{h}_t(E) = 0$ and $\widehat{h}_t(j) = \widehat{h}_{t,j}$. The action selected by a fixed price is the lower envelope of the $k+1$ lines

$$C_t(a) + \lambda \widehat{h}_t(a), \qquad a \in \{E, 1, \dots, k\}.$$

With a deterministic tie-breaking rule, the action can change only when two of these lines cross.

Corollary 3 (Uniform certification over all prices)**.** *Fix a bounded interval* $[0, \lambda_{\max}]$ *and include the expert-only policy as an additional candidate. Let* Π_λ *be the fixed-price policy induced by* (14)–(16) *on the union of the target and calibration items. The number of distinct induced policies over* $\lambda \in [0, \lambda_{\max}]$ *is at most*

$$N_{\mathrm{br}} \le 1 + 2(T+m)\binom{k+1}{2}. \tag{34}$$

Therefore Theorem 1 holds uniformly over all $\lambda \in [0, \lambda_{\max}]$ *after replacing* $|\Lambda|$ *in* (21) *by* N_{br}*.*

Proof. For a fixed item, the selected action is constant between consecutive crossing points of pairs of lines $C_t(a) + \lambda \widehat{h}_t(a)$. There are at most $\binom{k+1}{2}$ pairwise crossings per item. Across $T+m$ target and calibration items, there are therefore at most

$$B = (T+m)\binom{k+1}{2}$$

crossing values, counting multiplicity. These crossing values partition $[0, \lambda_{\max}]$ into at most $B+1$ open intervals, and there are at most B exact crossing points. Within each open interval the entire vector of decisions on target and calibration items is fixed. At an exact crossing point, deterministic tie-breaking determines a single additional policy, which may or may not coincide with one of the neighboring interval policies. Thus the number of distinct induced policies is at most $2B+1$. Applying Theorem 1 to one representative price from each induced policy gives the result. □

D.2 Explicit Cost Accounting

This accounting is useful for implementation, but it is not central to the statistical guarantee. The main distinction is between deployment cost, development-label cost, and audit or correction cost.

Deployment labeling cost. For a fixed offline price λ, the average deployment cost on the target pool is

$$C_T^{\mathrm{dep}}(\lambda) = \frac{1}{T}\sum_{t=1}^{T}\left[D_t^\lambda c_{t,E} + (1 - D_t^\lambda) c_{t,j_t^\lambda}\right]. \tag{35}$$

This is the cost optimized in (22). If cheap predictions have already been precomputed at no marginal deployment cost, set $c_{t,j} = 0$. If calling a model has a per-token, per-image, or per-query cost, include that cost in $c_{t,j}$.

Offline development cost. If m_{train} expert labels are used to train $\widehat{h}$ and m_{cal} expert labels are used to certify the fixed price, the development cost is

$$C^{\text{dev}} = \sum_{i \in I_{\text{train}}} c_{i,E} + \sum_{i \in I_{\text{cal}}} c_{i,E} + \text{cheap-source costs incurred on development items.} \tag{36}$$

This cost may be treated as sunk if the labeled development data already exist, but it should be included when comparing end-to-end data-curation pipelines.

Audit-or-correction cost. In the offline setting, calibration labels play the role that audits play online. If calibration items are part of the target pool and are corrected to expert labels in the released dataset, then there is no duplicate expert cost for those items. The total realized end-to-end cost can be written as

$$C_T^{\text{total}}(\lambda) = \frac{1}{T} \sum_{t=1}^{T} \Big[(1 - D_t^{\lambda}) c_{t, j_t^{\lambda}} + \mathbf{1}\{D_t^{\lambda} \text{ or } t \in I_{\text{cal}}\} c_{t,E} \Big] + C^{\text{train}}/T, \tag{37}$$

where C^{train} denotes expert-labeling cost used only for proxy training and not for final target labels. This formula separates expert labels used for final deferral from expert labels used for statistical certification.

Online audit cost. For Algorithm 2, the conditional expected round-t cost under audit-and-correct is

$$\mathbb{E}[C_t \mid \mathcal{G}_t] = D_t c_{t,E} + (1 - D_t) \left(c_{t, j_t^{\star}} + p_t c_{t,E} \right).$$

This identity isolates the extra expert cost paid to obtain online feedback; it is not a cost-regret guarantee.

E Why Arbitrary Offline Labels Are Insufficient

Without a relationship between the offline labeled data and the target pool, no distribution-free guarantee is possible. Suppose an offline labeled dataset shows that a cheap source is always correct, while on the target pool the same source is always wrong. If the learner observes no target labels, the two worlds are indistinguishable from the target-pool covariates and cheap predictions alone. Thus a PAC guarantee must rely on randomized target-pool labels, an i.i.d. sampling assumption, a covariate-shift correction assumption, or a structural assumption on the loss proxy. This is the offline analogue of the online no-free-lunch issue under partial feedback.

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